



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI PERIYANDAVAR FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

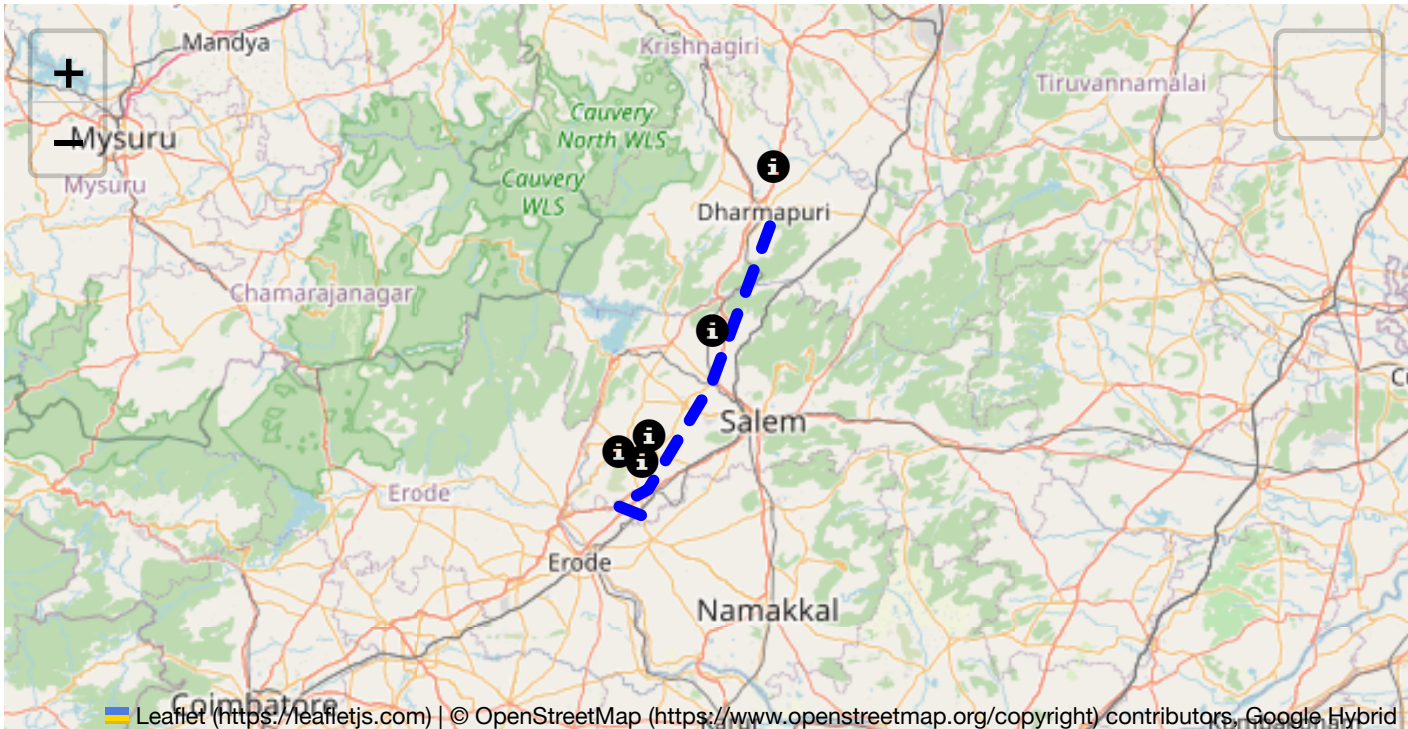
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

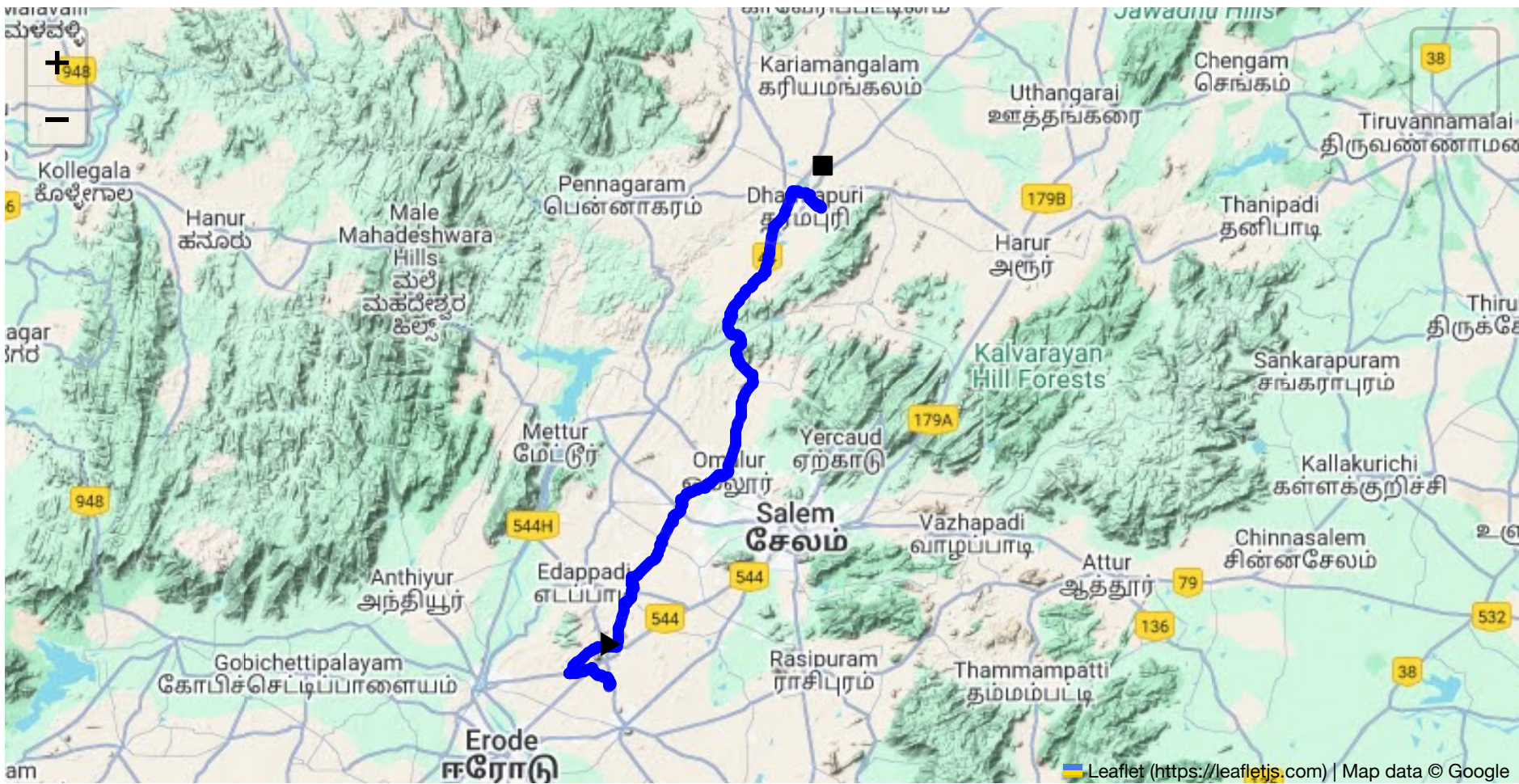
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 111.33 km
Estimated Duration: 2.4 hours
Adjusted Duration (Heavy Vehicle): 3.1 hours

Start: (11.4381, 77.8734)
End: (12.1171049, 78.1806317)



Welcome to the Journey Risk Management Study

Route Overview

The route covers approximately 111.33 kilometers, connecting Sangagiri to Annasagaram, passing through key waypoints like Padaiveedu, SH 86 through Ayveli, and Salem. The journey typically takes around 2.45 hours for heavy vehicles carrying hazardous materials.

1. Weather Conditions and Hazards

The region experiences a tropical climate, with significant rainfall from June to September due to the southwest monsoon. During monsoon seasons, waterlogging can be an issue, leading to reduced visibility and slippery roads. High temperatures during the summer months can also affect vehicle performance.

2. Traffic Patterns

Traffic congestion is common in Salem, especially during rush hours (8-10 AM and 5-7 PM). The road through Bhavani Main Road and SH 86 tend to have moderate traffic, which might slow down during peak times or festival seasons.

3. Road Quality and Infrastructure

The road quality is generally good on the highways, but certain stretches near rural areas or under construction zones can have potholes and uneven surfaces. Regular maintenance is advisable for trucks to handle these conditions.

4. Alternative Routes

In emergencies, detours through NH 44 towards Dharmapuri from Salem can be considered, though it might extend travel time. It's crucial to evaluate the nature and extent of the emergency before choosing

this route due to traffic and distance.

5. Local Regulations on Hazardous Materials

Tamil Nadu follows strict regulations for transporting hazardous materials. Carriers must comply with guidelines regarding vehicle signage, personnel training, and specific transit permissions through congested areas.

6. Historical Incidents

Data suggests occasional incidents involving heavy vehicles, primarily due to mechanical failures or loss of control on slippery roads during monsoons. However, there are no major historical incidents reported specifically for hazardous materials on this route.

7. Environmental Considerations

The route traverses relatively rural areas with significant agricultural land. Care must be taken to avoid any spills or leaks that could harm local ecosystems. The section through Salem requires particular attention due to urban density.

8. Communication Coverage

Most areas along the route receive good cellular network coverage, though some rural stretches may experience reduced signal strength, leading to occasional dead zones. Ensuring communication is crucial for real-time updates and emergency coordination.

9. Emergency Response Times

In urban areas like Salem, emergency response times are typically within 30-45 minutes. In rural stretches, it could extend up to 1-2 hours. It is vital to report incidents immediately to minimize delay in response.

Overall Risk Assessment

The route is moderately challenging for heavy vehicles, especially during adverse weather conditions and peak traffic periods. Regular updates on weather forecasts, road conditions, and awareness of alternative routes can mitigate risks. Compliance with local regulations and preparedness for communication issues are key to ensuring safe transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45853, 77.85139	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
5	Turn	Medium	11.45906, 77.85132	30 KM/Hr
6	Turn	High	11.46307, 77.84941	15 KM/Hr
7	Turn	High	11.45542, 77.81725	15 KM/Hr
8	Turn	High	11.45636, 77.81678	15 KM/Hr
9	Turn	Medium	11.49383, 77.88501	30 KM/Hr
10	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
11	Turn	High	11.49469, 77.88488	15 KM/Hr
12	Turn	High	11.55618, 77.89789	15 KM/Hr
13	Turn	Medium	11.55629, 77.89888	30 KM/Hr
14	Turn	Medium	11.58228, 77.90841	30 KM/Hr
15	Turn	Medium	11.58290, 77.90783	30 KM/Hr
16	Turn	High	11.58340, 77.90637	15 KM/Hr
17	Turn	Medium	11.63832, 77.94818	30 KM/Hr
18	Turn	Medium	11.64161, 77.95177	30 KM/Hr
19	Turn	Medium	11.64226, 77.95200	30 KM/Hr
20	Turn	Medium	11.64716, 77.95118	30 KM/Hr
21	Turn	High	11.67868, 77.96836	15 KM/Hr
22	Turn	Medium	11.70077, 77.97957	30 KM/Hr
23	Turn	High	11.70223, 77.97860	15 KM/Hr
24	Turn	Medium	11.71402, 77.99586	30 KM/Hr
25	Turn	Medium	11.71918, 78.01127	30 KM/Hr
26	Turn	Medium	11.72565, 78.01770	30 KM/Hr
27	Turn	Medium	11.73769, 78.03451	30 KM/Hr
28	Turn	Medium	11.73785, 78.03551	30 KM/Hr
29	Turn	High	11.73651, 78.03871	15 KM/Hr
30	Turn	Medium	11.74271, 78.04717	30 KM/Hr
31	Turn	Medium	11.78017, 78.05790	30 KM/Hr
32	Turn	Medium	11.86953, 78.08386	30 KM/Hr
33	Turn	Medium	11.88019, 78.08158	30 KM/Hr
34	Turn	Medium	11.91282, 78.05854	30 KM/Hr
35	Turn	Medium	11.93757, 78.05319	30 KM/Hr
36	Turn	Medium	11.95816, 78.04976	30 KM/Hr
37	Turn	Medium	11.95870, 78.04990	30 KM/Hr
38	Turn	High	11.96031, 78.04945	15 KM/Hr
39	Turn	Medium	11.96291, 78.05131	30 KM/Hr
40	Turn	Medium	11.96390, 78.05138	30 KM/Hr
41	Turn	Medium	11.96453, 78.05092	30 KM/Hr
42	Turn	High	11.96562, 78.04928	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
43	Turn	High	11.96667, 78.04934	15 KM/Hr
44	Turn	Medium	11.96775, 78.05056	30 KM/Hr
45	Turn	High	12.14051, 78.13980	15 KM/Hr
46	Turn	High	12.13658, 78.16179	15 KM/Hr
47	Turn	High	12.13651, 78.16182	15 KM/Hr
48	Turn	High	12.13576, 78.16498	15 KM/Hr
49	Turn	Medium	12.13083, 78.16460	30 KM/Hr
50	Turn	High	12.13039, 78.16472	15 KM/Hr
51	Turn	High	12.13010, 78.16585	15 KM/Hr
52	Turn	Medium	12.12886, 78.16640	30 KM/Hr
53	Turn	High	12.12824, 78.16754	15 KM/Hr
54	Turn	High	12.12743, 78.16755	15 KM/Hr
55	Turn	Medium	12.12592, 78.16942	30 KM/Hr
56	Turn	High	12.12584, 78.17033	15 KM/Hr
57	Turn	Medium	12.12521, 78.17040	30 KM/Hr
58	Turn	Medium	12.12492, 78.17069	30 KM/Hr
59	Turn	High	12.12426, 78.17070	15 KM/Hr
60	Turn	Medium	12.12316, 78.17441	30 KM/Hr
61	Turn	Medium	12.12220, 78.17521	30 KM/Hr
62	Turn	High	12.12162, 78.17532	15 KM/Hr
63	Turn	Medium	12.11919, 78.17859	30 KM/Hr
64	Turn	Medium	12.11866, 78.18033	30 KM/Hr
65	Turn	Medium	12.11816, 78.18062	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
11	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
12	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
13	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
16	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
21	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
25	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
26	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
29	hospital	Vijaya Hospital	12.1399758, 78.1532115	30 km/h	Medium
30	clinic	Bee Tee Clinic	12.138889, 78.1576748	30 km/h	Medium
31	clinic	Sheela Clinic	12.138725, 78.1573436	30 km/h	Medium
32	clinic	Suguna Clinic	12.1386752, 78.1575468	30 km/h	Medium
33	clinic	Bharani Neurologist	12.135969, 78.16255	30 km/h	Medium
34	hospital	Pragadeesh Hospital	12.135284, 78.1591867	30 km/h	Medium
35	hospital	Kamatchi Hospital	12.1350608, 78.1594841	30 km/h	Medium
36	hospital	Navina Hospital	12.1367521, 78.1610296	30 km/h	Medium
37	clinic	Malar Siddha Clinic	12.1350485, 78.159226	30 km/h	Medium
38	hospital	Kamala Nehru Hospital	12.13404, 78.1633476	30 km/h	Medium
39	hospital	D.N.V Hospital	12.1351328, 78.162869	30 km/h	Medium
40	clinic	Ragavendra Physiotherapy Clinic	12.135569, 78.1635247	30 km/h	Medium
41	clinic	Govindaraj Clinic	12.1344914, 78.1642372	30 km/h	Medium
42	hospital	Sri Priya Hospital	12.135347, 78.164391	30 km/h	Medium
43	hospital	Vasan Eye Care-Dharmapuri	12.1326054, 78.1630215	30 km/h	Medium
44	hospital	Supa Hospital	12.1325709, 78.1625215	30 km/h	Medium
45	hospital	Raaj Ortho Care Hospital	12.1325581, 78.163019	30 km/h	Medium
46	hospital	Jyothi Laser Eye Hospital	12.1324296, 78.1643197	30 km/h	Medium
47	hospital	Shyam Sundar Hospital	12.1311917, 78.1638458	30 km/h	Medium
48	clinic	Dr. Asokan Skin and ENT Care	12.1321209, 78.1639804	30 km/h	Medium
49	hospital	Nagaraj Hospital	12.1313002, 78.1637853	30 km/h	Medium
50	clinic	Sri Dhaya Clinic	12.1314423, 78.1622991	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
51	hospital	Kumudha Hospital	12.1324596, 78.1620599	30 km/h	Medium
52	clinic	Sri Dhahalakshmi Child Care Centre	12.1308627, 78.1640471	30 km/h	Medium
53	clinic	Sugam Polyclinic	12.1323296, 78.1629319	30 km/h	Medium
54	hospital	Sree Dhanalakshmi Hospital	12.1323769, 78.1631872	30 km/h	Medium
55	hospital	K.S. Hospital	12.1324134, 78.1620136	30 km/h	Medium
56	hospital	P.M.K. Hospital	12.131546, 78.1619069	30 km/h	Medium
57	hospital	Gowri Eye Hospital	12.1327566, 78.1618526	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49383, 77.88501



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55618, 77.89789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58340, 77.90637



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63832, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64161, 77.95177



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64226, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64716, 77.95118



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67868, 77.96836



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70077, 77.97957



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70223, 77.97860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71402, 77.99586



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71918, 78.01127



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72565, 78.01770



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73769, 78.03451



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73785, 78.03551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74271, 78.04717



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78017, 78.05790



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86953, 78.08386



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88019, 78.08158



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91282, 78.05854



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93757, 78.05319



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95816, 78.04976



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95870, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96031, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96291, 78.05131



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96390, 78.05138



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96453, 78.05092



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96562, 78.04928



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96667, 78.04934



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96775, 78.05056



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14051, 78.13980



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.13658, 78.16179



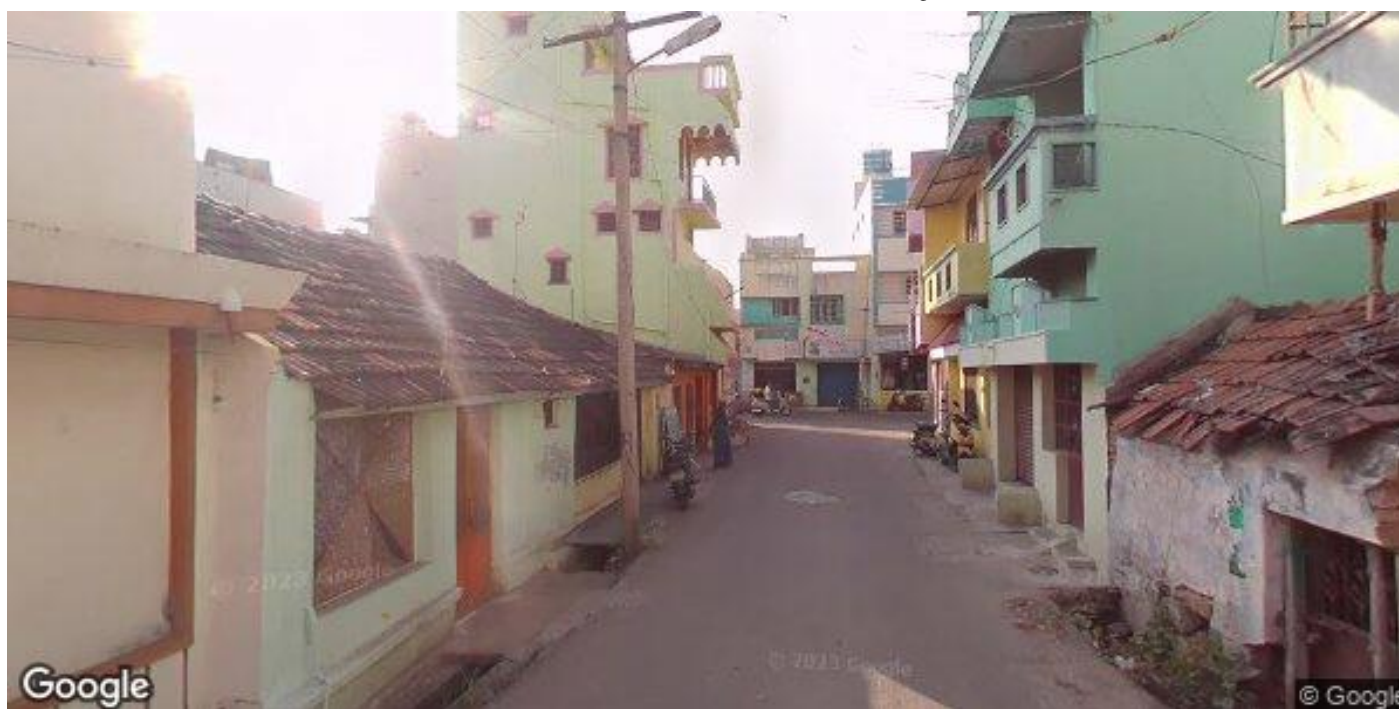
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.13651, 78.16182



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.13576, 78.16498



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.13083, 78.16460



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.13039, 78.16472



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.13010, 78.16585



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12886, 78.16640



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.12824, 78.16754



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.12743, 78.16755



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12592, 78.16942



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.12584, 78.17033



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12521, 78.17040



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12492, 78.17069



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.12426, 78.17070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12316, 78.17441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12220, 78.17521



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.12162, 78.17532



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11919, 78.17859



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11866, 78.18033



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11816, 78.18062
